06/01/24 M/S

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[22CSE141]

THIRD SEMESTER B.Tech EXAMINATION, DECEMBER - 2023

SOFTWARE ENGINEERING

Time: 3 Hours Maximum **Marks: 100**

Instructions: i. Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

		PART-A	**5 X 2=10**		
1.	**a.**	Differentiate Software engineering and System engineering.	L2	1	**02**
	b.	If you want to develop a word processing software product, what process models will you choose? Justify your answer.	L2	1	**02**
	c.	Write the fundamental activities that are involved in software development process?	L4	4	**02**
	d.	Compare Testing and Debugging.	L3	5	**02**
	e.	What are the two basic ways to estimate the size of a project.	L3	6	**02**
		PART-B	**6 X 5=30**		
2.	**a.**	With a neat diagram, Explain the waterfall model.	L2	1	**05**
		OR			
	b.	What are the attributes of good software? Explain the key challenges faced in software engineering.	L2	1	**05**
3.	**a.**	Write a note on Notation used for writing systems requirements.	L2	2	**05**
		OR			
	b.	Organize and Explain structure of software requirements documents.	L2	2	**05**
4.	**a.**	Explain modular design with an example.	L3	3	**05**
		OR			
	b.	Define architectural styles and list the architectural styles.	L4	4	**05**
5.	**a.**	Elaborate the purpose of program inspection.	L4	4	**05**
		OR			
	b.	State the various implementation issues of software development.	L4	4	**05**
6.	**a.**	Explain the three different types of testing carried out during software development.	L3	5	**05**
		OR			
	b.	Explain validation testing with an example.	L3	5	**05**
7.	**a.**	Give a complete template for SRS.	L2	2	**05**
		OR			

	b.	Discuss the desirable characteristics of feasibility studies.	L2	2	05
		PART-C	**6 X 10=60**		
8.	a.	Justify "Design is not coding and coding is not design".	L2	1	10
		OR			
	b.	Identify and explain principal stages of spiral model with a neat diagram. Mention its pros and cons.	L2	1	10
9.	a.	Differentiate functional and non-functional requirements in software and explain.	L2	2	10
		OR			
	b.	Explain the components of SRS.	L2	2	10
10.	a.	Give the sketch of requirements engineering process and explain the different stages?	L3	3	10
		OR			
	b.	Indicate why requirements validation is needed. Discuss different checks to be carried out during this requirement validation process.	L3	3	10
11.	a.	What are the different types of user testing and explain the six stages of acceptance testing process with neat diagram.	L3	5	10
		OR			
	b.	Write short note on: **i.** White-box Testing **ii.** Black-box Testing **iii.** Validation Testing	L3	5	10
12.	a.	Explain Risk management concepts in detail.	L3	6	10
		OR			
	b.	How would you define software quality? Briefly discuss the quality attributes.	L3	6	10
13.	a.	Describe the following with respect to project plan development: **i.** Sections of project plan **ii.** Project scheduling	L3	6	10
		OR			
	b.	With a neat diagram, explain the COCOMO estimation model.	L3	6	10